

Chapter 16.1: Scalar Line Integrals

V1: Line Integrals. I can set up and evaluate scalar and vector field line integrals in two and three dimensions.

A2: Integral Applications. I can use multiple integrals to solve physical problems, such as finding area, average value, volume, or the mass or center of mass of a lamina or solid. I can interpret mass, center of mass, work, flow, circulation, flux, and surface area in terms of line and/or surface integrals, as appropriate.

Mechanics

- For each curve, find a parametrisation of the curve with the specified orientation.
 - The line segment in $\mathbb{R}^3$ from $(0, 1, -2)$ to $(3, -1, 2)$.
 - The line segment in $\mathbb{R}^3$ from $(3, -1, 2)$ to $(0, 1, -2)$.
 - The circle of radius 3 in $\mathbb{R}^2$ centered at the origin, beginning at the point $(0, -3)$ and proceeding clockwise around the circle.
 - The ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ in $\mathbb{R}^2$ beginning at the point $(a, 0)$, $a > 0$ and proceeding counterclockwise around the circle.
 - In $\mathbb{R}^2$, the graph of *any* smooth function $y = f(x)$ from $(x_0, f(x_0))$ to $(x_1, f(x_1))$, where $x_0 \leq x_1$.
 - In $\mathbb{R}^2$, the graph of *any* smooth function $x = g(y)$ from $(g(y_0), y_0)$ to $(g(y_1), y_1)$, where $y_0 \leq y_1$.
- Find the line integral of $f(x, y) = \sqrt{4x + 1}$ over C where C is the part of the curve $x = y^2$ from the point $(4, -2)$ to $(1, 1)$.
- Find the line integral of $f(x, y, z) = \sqrt{x^2 + y^2}$ over the curve $\mathbf{r}(t) = (-4 \sin t)\mathbf{i} + (4 \cos t)\mathbf{j} + 3t\mathbf{k}$, where $t \in [0, 2\pi]$.

- Evaluate the line integral

$$\int_C (x^2 - yz) ds$$

ere C is parameterized by $\mathbf{r}_1(t) = (2t, 5t + 1, -t)$ for $t \in [0, 10]$. Recalculate the integral using the parameterization $\mathbf{r}_2(t) = (4t, 10t + 2, -2t)$ for $t \in [0, 5]$. What do you notice? Is this a coincidence? Summarize your findings by filling in the blank:

Scalar line integrals are independent of _____ (so long as the path is traversed exactly once).

Applications

5. A awkwardly shaped animal enclosure involves a fence with base parametrised by the closed curve $\langle t^2 - 1, t(t^2 - 1) \rangle$ for $-1 \leq t \leq 1$ and height equal to $\sin(xy) + 2$. Set up a line integral to compute the surface area of the fence.
6. Suppose you are designing a custom tablecloth for a circular dinner table modelled by $x^2 + y^2 \leq 1$. Your customer would like the cloth to drape down $f(x, y) = -x^2 - y^2 + xy - 4$ units off of the boundary of the table. Compute the total area of fabric needed for the tablecloth.
7. Let C a thin wire parameterized by

$$\mathbf{r}(t) = (e^t \cos t)\mathbf{i} + (e^t \sin t)\mathbf{j} + e^t\mathbf{k}, \quad t \in [0, \pi].$$

Find the mass of C if the density along C is $\delta(x, y, z) = z^{-1}$.

Extensions

8. Consider the line integral

$$\int_{C_k} (x^2 + y^2 - z) ds$$

where C_k is the part of the helix $\langle \cos t, \sin t, t \rangle$, for $0 \leq t \leq k$. Find the value of k which maximizes the integral.